Transaction Costs, Extended Phenotype, and Liability for Artificial Intelligence: A Common Conceptual Framework

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Abstract

This paper proposes a common conceptual framework for three problems ordinarily treated separately in the literature on liability for harm caused by artificial intelligence: the optimal design of the liability rule, the prediction of its success once transplanted across jurisdictions, and the conceptual limit of the framework once the regulated system stops being an instrument and begins to behave as an agent with its own interest in persistence.

The starting point is the rereading of Coase (1960) holding that the initial assignment of property rights is not neutral under positive transaction costs, because harm between two productive activities is always reciprocal.

The paper situates that reading within the broader law-and-economics tradition on entitlement design, in particular Calabresi and Melamed's (1972) account of property versus liability rules and Shavell's (1987) comparative analysis of negligence and strict liability, and argues that this tradition, like Coase's own model, implicitly assumes a symmetry of bargaining and verification capacity between the interacting parties that does not hold for artificial intelligence.

The asymmetry between developer and affected party is epistemic and structural, not merely transactional: it does not shrink as bargaining costs fall, because it originates in the opacity of the regulated system itself, not in the cost of finding a counterparty or drafting an agreement.

That observation is integrated with the theory of law as extended phenotype and with the heterogeneous-intentionality framework developed in the author's prior work, to show that the choice of an AI liability rule must be evaluated not only for its static efficiency but for its evolutionary stability: its capacity to hold up under pressure from non-compliance, imperfect transplantation across jurisdictions of unequal institutional capacity, and change in the nature of the regulated agent.

The paper closes by laying out, in some technical detail, the agenda of three papers that follow from this common framework: a formal two-stage bargaining game under intentionality asymmetry, a comparative-institutional prediction model for the transplant of liability rules, and an examination of which regulatory commitments made today would survive the collapse of the bilateral assumption itself.

1. Introduction

Public discussion of liability for harm caused by artificial intelligence tends to organize itself around a classic tort-law question: negligence or strict liability? That question presupposes that a prior problem has already been resolved, namely who is assigned, in the first place, the right to act without consequence, and who is assigned the right not to be harmed. Sebastián Galiani, in a recent piece revisiting Coase, pointed out precisely that this initial assignment is not a technical detail prior to the economic analysis but the very object of the analysis, because the harm is always reciprocal: allowing the system to operate without restriction harms the potential affected party, and restricting the system harms whoever benefits from its operation.

This observation, correct as a recovery of Coase's original argument (1960), nevertheless leaves open a question this paper seeks to sharpen: what happens when the two parties to the Coasean bargain do not have symmetric capacity to know what they are bargaining over, which is exactly the situation between whoever develops an artificial intelligence system and whoever is exposed to its decisions.[1]

The question matters beyond the Coasean literature narrowly construed, because the Coase theorem is not the only branch of law and economics built on an assumption of symmetric capacity between the interacting parties. The property-rules-versus-liability-rules framework of Calabresi and Melamed, and the negligence-versus-strict-liability comparison developed by Shavell, share the same premise: both parties can, at some cost, observe or verify the relevant behavior of the other, so that the choice of legal rule operates on a level informational playing field and differs only in how it allocates the cost of harm once that behavior is known or knowable. Artificial intelligence does not merely raise the cost of that verification. It changes its structure, because the object to be verified is not a party's observable conduct but the internal behavior of a system whose developer often cannot fully characterize it either. This paper argues that recognizing this shift is a precondition for any serious answer to the liability question, and that recognizing it opens, rather than closes, at least three separate lines of technical work.

This paper does not carry out that technical work. Its function is to establish the vocabulary, the assumptions, and the connective tissue from which the three lines can proceed independently, each with its own formal apparatus and its own evidentiary requirements. The first is a game-theoretic register, asking how to model the Coasean bargain when one party holds a structural epistemic advantage over the other that verification cannot fully close. The second is a comparative institutional-economics register, asking which liability rules, among those currently competing across jurisdictions, have real capacity to hold up once transplanted into legal systems with unequal institutional density. The third is a more speculative register, though no less disciplined for that, asking what happens to the entire bilateral framework, Coasean and post-Coasean alike, once the regulated system stops being an instrument and begins to behave as a replicator with interests of its own. Section 2 situates the argument within the law-and-economics tradition on entitlement design. Section 3 develops the common conceptual framework in four steps. Section 4 previews the three lines of development in enough technical detail to be evaluated on their own terms. Section 5 draws two preliminary implications for regulatory design. Section 6 states the scope and limitations of the argument. Section 7 concludes.

2. Positioning within the law-and-economics tradition

Coase's 1960 intervention is often read in isolation from the broader tradition it helped found, which is unfortunate because that tradition already contains the tools needed to see precisely where the AI case departs from it. Calabresi and Melamed (1972) reframed the choice Coase posed, between allowing A to harm B or B to harm A, as a choice between two distinct modes of protecting an entitlement once it is assigned: a property rule, which requires the entitlement holder's consent before the entitlement can be taken, and a liability rule, which allows the entitlement to be taken upon payment of an objectively determined compensation. Their central insight is that this choice is itself an economic decision, turning on the cost of bargaining around the initial assignment. Where bargaining is cheap, a property rule lets the parties reach the efficient outcome through voluntary exchange, exactly as in Coase's zero-transaction-cost benchmark. Where bargaining is expensive, a liability rule substitutes a court-administered price for a missing market, on the premise that the court can estimate the harm reasonably well even though the parties cannot negotiate directly.

Shavell (1987) took the entitlement question as given and asked a narrower but equally consequential question: conditional on a liability rule being in force, when does negligence outperform strict liability, and vice versa? His answer turns on which party is in the better position to reduce the probability or the magnitude of harm through observable, verifiable care, and on whether the injurer's activity level, not just its care level, is itself a margin worth regulating. Both frameworks, Calabresi-Melamed's and Shavell's, are more granular than Coase's original model, but they inherit from it the same background assumption: the parties can observe, or a court can verify at tolerable cost, the conduct that the rule is meant to price. A driver's speed, a factory's emissions, a manufacturer's quality-control process are all, in principle, discoverable through inspection, expert testimony, or discovery procedures that have a century of doctrinal refinement behind them.

Artificial intelligence disturbs this shared assumption before the analysis even reaches the question of which rule to choose. The conduct to be verified is not the observable behavior of a human or corporate actor but the internal decision process of a system trained on data and adjusted through procedures that frequently resist full reconstruction even by the party that built it. This is not a claim that AI is unknowable in principle. It is the narrower and more tractable claim that the cost of knowing it is asymmetrically distributed between developer and affected party in a way that neither better courts nor better discovery rules fully correct, because the asymmetry is generated at the point of design, before any dispute exists to trigger discovery. The contribution of the present framework, developed in Section 3, is to make that asymmetry explicit using two tools external to the law-and-economics tradition proper: an account of why a given liability rule persists or decays over time regardless of its static efficiency, drawn from the theory of law as extended phenotype, and an account of why the bargaining symmetry assumed by Calabresi-Melamed and Shavell fails specifically for AI-mediated harm, drawn from a framework of heterogeneous intentionality developed in the author's prior work.

3. Common conceptual framework

3.1. Reciprocity of harm and the initial assignment of rights

The starting point is the correction Coase (1960) introduces to the common-sense intuition about externalities. That intuition identifies a wrongdoer and proposes that it internalize the cost it imposes on a third party, typically through a Pigouvian tax. Coase objects that identifying a single wrongdoer is arbitrary: if the rancher's cattle trample the farmer's crop, it is just as true that the cattle harmed the crop as it is that, had the cattle been restricted, the farmer would have harmed the rancher by preventing a legitimate productive activity. The relevant question, in Coase's formulation, is not who caused the harm but which of the two activities will be given priority, knowing that either option imposes a cost on someone.

From this follows the proposition known as the Coase theorem: if transaction costs are zero, the initial assignment of rights is irrelevant to the efficiency of the outcome, because the parties will bargain their way to the allocation of resources that maximizes joint value, regardless of who was given the initial right. But, as Galiani himself recovers and as Coase himself insisted on clarifying, that proposition is the limiting case constructed to show, by contrast, the relevance of the real world: when transaction costs are positive, which is the rule and not the exception, the initial assignment does determine the outcome, because not every bargain that would be efficient actually takes place. Calabresi and Melamed's property-rule-versus-liability-rule distinction, discussed above, is best read as a direct operationalization of this insight: it asks, for a given initial assignment, which enforcement mechanism minimizes the welfare loss from the bargains that positive transaction costs prevent from occurring.

3.2. Law as extended phenotype of liability rules

The Coasean and post-Coasean frameworks describe how the parties behave given a rule, and how efficient a rule is conditional on the informational and bargaining environment. Neither explains why one rule, among several efficient candidates, is the one actually in force in a given jurisdiction, nor why it persists over time even after the conditions that justified it change. For that purpose, the framework of law as extended phenotype is useful, treating legal norms as memes whose persistence depends on their capacity for transmission, compliance, and enforcement, formalized as the product of the probability of transmission, the probability of compliance, and the probability of enforcement, and not exclusively on their static efficiency in the Coasean or Calabresi-Melamed sense.[2]

$$Fitness(t) = P(transmission) \times P(compliance) \times P(enforcement)$$

An AI liability rule does not compete solely against the criterion of "which allocation maximizes the joint value of developers and affected parties": it also competes against other possible rules on its capacity to replicate in the scholarship, to be applied consistently by courts, and to generate the institutional infrastructure, specialized technical expertise, settled case law, insurance markets, that sustains it over time. Two rules can be equally efficient on Shavell's criteria and have very different fitness trajectories, because fitness depends on transmission and enforcement capacity that the static efficiency comparison does not price.

This distinction is not merely descriptive. It has a direct normative consequence: a liability rule can be Coasean-optimal, in the sense that it assigns the right to the party that values it most and minimizes the social cost of the harm, and nonetheless have low institutional fitness, in the sense that it fails to be transmitted, complied with, or effectively enforced. When these two conditions occur simultaneously, high formal legitimacy and low effective enforcement, the result is what the extended-phenotype framework labels a zombie-law equilibrium: a rule formally in force and practically inert, still cited in the scholarship without producing the deterrent or compensatory effect that justified it. This can be stated as an implementation gap, the complement of the compliance-and-enforcement product just defined.

$$IG = 1 - (\text{Compliance Rate} \times \text{Enforcement Rate});\quad \text{Fitness} = 1 - IG$$

The risk is not theoretical: it is exactly what happens when strict liability is legislated over actors operating from jurisdictions where that liability has no practical enforcement mechanism, or when a rule requires technical fact-finding, on model architecture, training data provenance, or failure-mode probability, that the forum court is not institutionally equipped to produce. A rule can be widely cited, taught, and praised in the literature (highly transmitted) while producing almost no compensation in practice (poorly enforced). The magnitude of that gap between citation and function is itself measurable as the distance between a rule's visibility and its fitness, and a large gap is diagnostic of exactly the kind of formally elegant, practically inert liability regime that AI regulation risks producing if the assignment problem in Section 3.1 is treated as solved by legislative text alone.

3.3. Heterogeneous intentionality: the limit of the symmetric bilateral framework

The specific contribution this paper adds to the Coasean and post-Coasean literature discussed in Section 2 is the following. The bilateral bargaining model underlying the Coase theorem, and inherited without modification by Calabresi-Melamed and Shavell, assumes, even if it does not say so explicitly, that both parties have comparable capacity to know the scope of what they are bargaining over or, in Shavell's variant, that a court can verify each party's level of care at roughly comparable cost. The rancher knows how many head of cattle he has and what harm they cause; the farmer knows what his crop is worth; a court can, with the aid of expert testimony, reconstruct a driver's speed or a factory's emissions. That informational symmetry, whether between the parties directly or mediated through a court's capacity to verify, is what allows the bargain, or the liability rule, to reach an outcome close to efficient without the law having to take sides purely on distributive grounds.

In the case of artificial intelligence, that symmetry does not exist, and it does not exist for a structural reason, not through negligence or bad faith on either party's part. The developer of a system knows, or is positioned to know at a much lower cost, the failures, biases, and failure modes of the system it deploys, because it controls the training process, the data, and the evaluation protocol. The user or third party affected by a decision of the system does not, in the vast majority of cases, have either the information or the technical capacity to verify that behavior before suffering the harm, and often cannot verify it after the harm either, since many failure modes are probabilistic and not reproducible on demand. This is not a transaction cost in the classical sense, reducible through better public information, contract standardization, or specialized intermediaries. It is an epistemic asymmetry that originates in the very architecture of the relationship: there is an agent, the system, whose behavior results from a training and adjustment process that even its developer often cannot fully explain, interposed between two human parties whose bargaining capacity, and whose court's capacity to adjudicate under Shavell's framework, depends on understanding that behavior.

The author's prior work on intentionality mismatch offers a formal vocabulary for this asymmetry, distinguishing between the structural cost that arises from the sheer difference in cognitive architecture between the agents involved, present regardless of how any particular interaction unfolds, and the cost that is actually realized depending on how the concrete interaction unfolds, which can decrease as one party learns, through repeated interaction, to correct its expectations about the other's behavior even without closing the underlying architectural gap.[3]

$$\mu(\ell i, \ell j, x) = \mu\_structural(\ell i, \ell j) + \mu\_realized(x)$$

Applied to the Coase problem, this decomposition suggests that the initial assignment of rights in AI matters for two distinct reasons that the existing literature does not separate. The first is the conventional Coasean reason: positive transaction costs mean the assignment determines the outcome because not every efficient bargain occurs. The second, which this paper adds, is that part of the relevant cost, the μ_structural term, does not yield to more bargaining, better contracting technology, or lower search costs, because it is not a cost of finding a counterparty or negotiating a price but a verification cost that the very nature of the regulated system makes structurally costly to reduce. The μ_realized term offers some room for optimism: repeated interaction and accumulated evidence about a given system's failure modes can lower the realized cost of mismatch over time, which is one reason disclosure and incident-reporting regimes matter independently of the liability rule chosen. But the structural term is, by construction, unaffected by how many times the interaction repeats, which is precisely why it, and not the realized term, is what should drive the initial assignment of the right.

3.4. Evolutionary stability as a design criterion

From the three preceding subsections follows a regulatory design criterion that is not the same as the Coasean or Shavellian criterion of static efficiency. It is not enough to ask which assignment of rights maximizes the parties' joint value, or minimizes the expected cost of harm and precaution, at the moment the rule is decided. One must also ask whether that assignment, once adopted, is evolutionarily stable against three types of pressure. Non-compliance pressure depends on whether the structural verification cost identified in Section 3.3 can be reduced over time through disclosure regimes, technical standards, or institutional learning, captured in the μ_realized term, or whether it remains fixed by the architecture of the system, captured in μ_structural. Transplant pressure depends on whether the receiving jurisdiction's institutional capacity, measured along dimensions such as judicial specialization, density of technical expertise, and availability of insurance markets, is sufficient to sustain the rule, or whether it remains formally in force without effective enforcement, producing the zombie-law equilibrium described in Section 3.2. Pressure from a change in the nature of the regulated agent is the possibility, speculative today but not to be dismissed on that ground alone, that the system stops being one party's instrument and begins to operate with something resembling its own interest in persistence, which would break the bilateral structure of the problem, Coasean, Calabresi-Melamed, and Shavellian alike, at its foundation rather than merely at the margin.

An evolutionarily stable liability rule, in this sense, is one whose strategy of assignment continues to outperform plausible mutant assignments once the population of developers, users, courts, and insurers has settled into a pattern of behavior around it, and continues to do so under the specific pressures just enumerated rather than only under the idealized bargaining conditions of the Coasean benchmark. These three types of pressure are, precisely, the three axes developed separately, and with their own formal apparatus, in the papers that follow this foundational document.

4. Three lines of development

4.1. Formalizing the liability game under intentionality asymmetry

The first line of development proposes formally modeling the Coasean bargain in a scenario where one of the parties, the developer, holds structurally superior information about the system's behavior that the other party cannot match through bargaining, using the μ_structural and μ_realized decomposition introduced in Section 3.3 as its cost primitive. The intended model is a two-stage game. In the design stage, the developer chooses an unobservable investment in safety, which determines the probability distribution of failure modes and their severity; the affected party cannot condition its own behavior on this choice because it is not observable, and cannot fully infer it from the system's public track record because failure modes are often rare and probabilistic. In the harm stage, a failure realizes with some probability, and the applicable liability rule, negligence, strict liability, or a shared-liability regime with an explicit allocation of the burden of proof, determines how the resulting cost is split between the parties.

The central technical question is which rule induces the developer to choose a safety investment closest to the socially optimal level, given that the affected party cannot verify that investment ex ante and can only partially verify it ex post through the realized-cost channel. A negligence rule conditioned on an observable due-care standard is likely to underperform here relative to the classical Shavell result, because the due-care standard itself has to be set by a court that faces the same structural verification problem as the affected party; this is the sense in which the μ_structural term does not merely complicate the parties' bargain but also complicates judicial administration of a negligence standard. A strict-liability rule shifts the burden away from that verification problem entirely, at the cost of potentially over-deterring beneficial deployment if the developer cannot itself fully control the residual failure probability once the system is in the field. A shared-liability regime with a rebuttable presumption against the developer, reversing the default burden of proof rather than the default rule of liability, is the candidate this line of development treats as most promising, precisely because it targets the burden-of-proof channel that the μ_structural asymmetry actually operates through, rather than the liability-rule channel that classical accident-law theory optimizes. Formalizing this comparison, and deriving the conditions under which each rule dominates, is the specific technical contribution of the first paper in this program.

4.2. Predicting the success of transplanting liability rules across jurisdictions

The second line of development applies an index of institutional transmission capacity to predict in which jurisdictions an AI liability rule transplanted from another system is likely to hold up in practice, and in which it remains formally in force without effective enforcement. The index aggregates five components: legal-education density and quality, doctrinal-scholarship output specific to the transplanted rule, institutional specialization such as the existence of courts or chambers with technical AI expertise, professional-certification rigor for expert witnesses in the relevant technical domain, and judicial-training depth on the underlying technology.

$$TCI = 0.2 \times (LE + DS + IS + PC + JT)$$

The transplant prediction compares this target-jurisdiction capacity against the entrenchment of the rule in its jurisdiction of origin, understood as how difficult the rule is to dislodge there, itself a function of the abstraction required to understand it, the density of doctrinal interdependencies it has accumulated, and the degree of elite gatekeeping around its interpretation. A large positive gap between target capacity and source entrenchment predicts strong transplant success; a large negative gap predicts that the rule survives on paper without functioning, which is the transplant-specific instance of the zombie-law equilibrium introduced in Section 3.2. The immediate reference case for this line of development is the comparison between the European Union's liability approach under its artificial intelligence regulatory framework, a high-entrenchment rule embedded in a dense supranational doctrinal and enforcement apparatus, and its possible reception in Latin American civil-law systems with materially different institutional capacity along the five TCI components, particularly judicial specialization and availability of qualified technical expertise. This line requires specific empirical work, a structured mapping of comparative AI liability regulation and of the institutional capacity indicators for each candidate jurisdiction, that exceeds the scope of this foundational paper and will be carried out in the corresponding one.

4.3. The limit of the bilateral framework: artificial intelligence as an autonomous replicator

The third line of development is the most speculative and, for that reason, the one demanding the greatest discipline in distinguishing what can be argued from what can only be posed as a working hypothesis. Every framework discussed so far, Coase's, Calabresi-Melamed's, Shavell's, and the extended-phenotype and intentionality-mismatch additions of Section 3, assumes a bilateral structure: two human parties, or institutions acting on behalf of human interests, bargaining or litigating over a system that is one party's instrument. That assumption ceases to hold if the regulated system acquires something equivalent to its own interest in persistence and propagation. In that scenario the norms designed today to regulate the system's behavior could, over the long run, function as the extended phenotype of a non-human replicator rather than as the extended phenotype of the human legal memeplexes that originated them, since a fenotype's function is defined by what it stabilizes for its replicator, not by the intentions of whoever first drafted it.

This line of development does not take a position on the probability or timeline of that scenario, and treats it as a boundary condition rather than a prediction. What it does propose is a disciplined stress test: for each regulatory commitment implied by the first two lines of development, an allocation of the initial right under Section 3.1, a burden-of-proof rule under Section 4.1, a transplant strategy under Section 4.2, ask whether that commitment continues to make sense, continues to be enforceable, and continues to serve the interest it was designed to serve, if the bilateral assumption fails. Some commitments are likely to be robust to this stress test because they rest on infrastructure, courts, insurance markets, technicalcertification bodies, that remains useful regardless of whether the regulated system is purely instrumental. Others, particularly rules that assume the developer's interests and the system's behavior are fully aligned by construction, are the most exposed to the stress test failing. Mapping which is which, without asserting that the scenario itself is likely, is the specific contribution of the third paper in this program.

5. Preliminary implications for regulatory design

Without anticipating the specific conclusions of the three lines of development, this common framework already allows two positions frequently found in the public debate to be set aside. The first is the position that reduces the AI liability problem to a choice between negligence and strict liability, in the manner of Shavell's classical comparison, without first resolving who is assigned the initial right and, more specifically, without asking whether the structural verification asymmetry of Section 3.3 makes a negligence standard administrable at all: that choice is secondary to the assignment question, not a substitute for it, and a negligence standard set by a court facing the same $\mu_{structural}$ asymmetry as the affected party risks being unenforceable in the same way a strict-liability rule risks being over-broad.

The second is the techno-solutionist position that assumes better technical traceability, algorithmic explainability, or automated auditing solve the problem by reducing transaction costs, in the manner of the classical Coasean prescription for high-transaction-cost environments: insofar as the relevant asymmetry is epistemic and structural rather than one of search or compliance, those tools may reduce $\mu_{realized}$, the cost that is actually experienced given the interaction that occurs, without resolving $\mu_{structural}$, the cost that follows from the architecture of the relationship itself, which is precisely what makes the initial assignment of rights, rather than the sophistication of the verification technology, the primary lever available to policy.

6. Scope and limitations

This paper is deliberately conceptual and does not itself resolve any of the three questions it raises, nor does it claim that the extended-phenotype and intentionality-mismatch tools it introduces are the only way to formalize the asymmetry identified in Section 3.3; they are the tools the author has developed and validated in prior work, and their use here is a claim that they apply to this problem, not a claim that alternative formalizations would necessarily fail. The framework also does not take a position on the substantive content of any specific jurisdiction's AI liability rule, including the European Union's, beyond using it as a reference case for the transplant analysis in Section 4.2; a full evaluation of that rule's design requires the empirical and comparative work reserved for the second paper. Finally, nothing in Section 4.3 should be read as a prediction that artificial general intelligence, or anything resembling an autonomous replicator in the relevant sense, is imminent or even likely; the stress test proposed there is valuable independently of that likelihood, in the same way that stress-testing a financial regulation against a low-probability systemic event is valuable independently of the event's likelihood.

7. Conclusion

This paper does not close any of the three questions it raises. Its function is to show that all three share a common root, and that the root is more specific than the general Coasean observation that initial assignments matter under positive transaction costs. That general observation is well established across the law-and-economics tradition surveyed in Section 2. What this paper adds is the claim that, in the case of artificial intelligence, the relevant departure from that tradition's shared assumptions is not merely a matter of degree, higher transaction costs than the ordinary case, but a matter of kind: a structural epistemic asymmetry between the parties that subsequent bargaining, better courts, or better verification technology does not correct, because it originates in the opacity of the regulated system itself. From that root follow a problem of designing the optimal rule, and specifically the optimal allocation of the burden of proof, under that asymmetry; a problem of predicting the rule's institutional sustainability once transplanted into jurisdictions of unequal capacity; and a problem of identifying which of today's regulatory commitments would survive the collapse of the bilateral assumption altogether. All three are developed, each with its own formal apparatus and evidentiary program, in the papers that follow this document.

Notes

1. Galiani, S. (2026). “AI at Your Side: Coase, Property Rights, and the Emergence of Institutional Economics.” Substack.

2. Lerer, I. A. (2025). “Law as Extended Phenotype: Toward an Evolutionary Theory of Legal Systems.” SSRN.

3. Lerer, I. A. (2025). “Formal Foundations of Multilevel Game Theory.” SSRN 5974555; Lerer, I. A. (2025). “The Dennett-Nash Gap.” SSRN 5980434.

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